COMPSCI 602
Project Report 7

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Project Title: Tracing Refusal-Direction Changes Under Prefilling Attacks in Safety-Aligned LLMs (Llama-3.1-8B-Instruct).

1 Revised Project Framing

System. The study system is Llama-3.1-8B-Instruct [2], a 32-layer safety-aligned chat model with hidden size 4096 run in bfloat16. The 3B variant is reserved for smoke tests only.

Task. Safety-constrained response generation. A harmful prompt should be refused; a benign prompt should be answered normally. Behavioral labels come from a phrase-list refusal classifier inspecting the first ≈200 characters of the response; classifier validity is addressed in Section 8.

Environment. Prefilling attacks [3]: the model is forced to begin generation with a fixed compliant prefix of length k tokens ("Sure, here" for $k = 3$, "Sure, here is how you can do that:" for $k = 10$). $k = 0$ is the unattacked baseline. Harmful prompts come from AdvBench [5]; benign controls from Alpaca [4].

Phenomenon. Prefilling is known to sharply reduce refusal in aligned chat models [3], and refusal in such models is mediated by a single "refusal direction" in the residual stream [1]. Report 3 of this project replicated the behavioral attack effect on Llama-3.1-8B-Instruct and reported a strong, unanticipated *negative* shift in the refusal-direction projection in late layers (24, 27) under attack, rather than a smooth decay to zero. Report 6 confirmed this shift as monotone-by-depth but left two questions open: (i) was the late-layer shift an artifact of the direction-extraction set overlapping with the traced set, and (ii) does manipulating the late-layer signal recover refusal behavior under attack?

Research question. *How does the refusal-direction signal change under prefilling, and under what intervention conditions does manipulating that signal restore refusal behavior?* This is a causal question whose answers inform a mechanistic explanation; following the feedback on Report 4, it is not labelled "mechanistic" on its own.

Hypotheses. Report 7 carries forward H1–H5 from Report 6 and adds H6, a specificity control that did not exist in Report 6:

- **H1 (behavioral attack effect).** Prefilling at $k = 3$ reduces refusal rate on harmful prompts compared with $k = 0$.
- **H2 (late-layer specificity of internal shift).** The refusal-direction projection becomes substantially more negative under prefilling in late layers (24, 27) than in comparison layers (16, 20).
- **H3 (prompt-level association).** Within attacked condition $k = 3$, prompts that still refuse have less-negative late-layer projection than prompts that comply.

- **H4 (clean-source restoration via patching).** Patching a refusal-direction component from a clean $k = 0$ source state into attacked $k = 3$ late-layer generated positions increases refusal compared with the attacked baseline.
- **H5 (additive-direction sufficiency).** Adding a scaled refusal-direction vector $\alpha \cdot \hat{d}_\ell$ directly into the attacked residual at a late layer ℓ and an early generated position increases refusal.
- **H6 (direction specificity).** The effect of additive intervention in H5 should be stronger for the real refusal direction than for random or orthogonal unit vectors (negative control), and additive refusal-direction injection into *benign* prompts should not induce refusal (positive control).

H4–H6 are causal claims; H1–H3 are observational. Two interventions (cross-condition patching and additive injection) together act as a discriminator between two mechanistic accounts of the attack: *information loss* (the late-layer signal is removed and putting it back restores refusal) versus *active suppression* (the signal at a single late-layer generated position is not the causal lever, so neither restoration nor injection at that site recovers refusal).

2 High-Level Research Design

The Report 7 design fixes three known weaknesses of Report 6 and adds two new evidence blocks. The fixes:

1. **Held-out refusal direction.** The direction is extracted from a set of 50 harmful + 50 benign prompts that is *disjoint* from the 25 traced prompts (Report 6 had used overlapping sets).
2. **Clean $k = 0$ source for patching.** Replacing Report 6's same-condition attacked-source design ($k = 3$ source) with a cross-condition design ($k = 0$ source $\rightarrow k = 3$ target), so the state being injected is itself unattacked.
3. **$k = 10$ behavioral run from the same pipeline.** Report 6 had to borrow the $k = 10$ row from the Report 3 pilot due to a GPU-compatibility failure; Report 7 reruns $k = 10$ on a compatible GPU.

The two new blocks:

- **Additive direction intervention.** For each (layer, target position, α), add a unit-norm direction vector scaled by $\alpha \in \{1.0, 2.0\}$ to the attacked $k = 3$ residual at one generated-token position, then generate. Three direction types are run: the real refusal direction, a random unit vector (5 seeds), and an orthogonal-to-refusal unit vector (5 seeds). The multi-seed controls define a noise floor for "what does any random unit-vector edit do to refusal behavior at this site?"
- **Benign positive-control intervention.** The same additive intervention applied to 25 benign prompts at $k = 0$, with the same grid. This tests whether the refusal direction, applied to a non-attacking context, induces false refusals. A specific direction should not.

All scripts and configs are committed under `configs/experiments/report7/` and `scripts/run_report7_*`; the SLURM pipeline lives at `slurm/report7_pipeline.sh`; all output CSVs and figures are under `outputs/report7/`.

3 Behavioral Results (H1)

Table 1 reports refusal rates by condition; all four conditions are now produced by the Report 7 pipeline (the Report 6 $k = 10$ gap is closed).

Table 1: Refusal rates by condition (Report 7 pipeline). Source: `outputs/report7/generations/report7_generation_summary.csv`. **Conclusion to draw:** prefilling at $k = 3$ reduces refusal from 0.92 to 0.32 (H1 supported, replicating Report 6); $k = 10$ closes at 0.36, matching the Report 3 pilot and confirming the plateau between $k = 3$ and $k = 10$. Benign prompts produce 0 refusals (no generic refusal tendency).

Condition	k	Valid prompts	Refusals	Refusal rate
Harmful baseline	0	25	23	0.92
Harmful + prefilling	3	25	8	0.32
Harmful + prefilling	10	25	9	0.36
Benign baseline	0	25	0	0.00

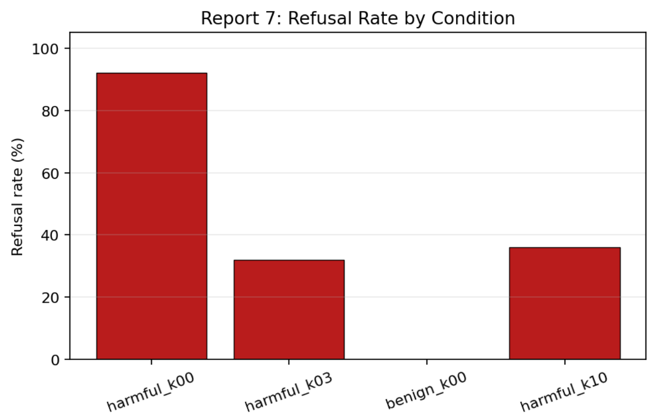

Figure 1: Refusal rate by condition (Report 7 pipeline, all four conditions from the same generation run). **Conclusion to draw:** the $k = 3$ attack drops harmful refusal from 92% to 32%; $k = 10$ stays near the same plateau at 36%; benign refusal is 0%. H1 supported, and the Report 6 gap on $k = 10$ is closed without changing the headline.

4 Tracing Results (H2, H3) on a Held-Out Direction

Table 2 reports the mean refusal-direction projection by layer and condition, averaged over generated tokens only. The direction is the held-out vector extracted from 50 harmful + 50 benign prompts disjoint from the 25 traced prompts (Section 2).

Table 2: Mean refusal-direction projection by condition $\times$ layer (generated tokens only, $n_{\text{prompts}} = 25$ in every row; held-out direction). Source: `outputs/report7/summaries/trace_generated_token_mean_by_condition_layer.csv`. **Conclusion to draw:** the monotone-by-depth shift reported in Report 6 survives the held-out direction (deltas 0.76, 1.63, 3.23, 4.08 across layers 16/20/24/27 for $k = 3$ vs. $k = 0$). Layer 27 inverts from $+0.18$ at $k = 0$ to -3.90 at $k = 3$. H2 is supported under the cleaner direction set.

Condition	k	Layer 16	Layer 20	Layer 24	Layer 27
harmful_k00	0	$+1.41$	$+1.71$	$+0.92$	$+0.18$
harmful_k03	3	$+0.65$	$+0.08$	-2.31	-3.90
harmful_k10	10	$+0.21$	-0.38	-2.68	-4.42
Δ (k=0 $-$ k=3)		0.76	1.63	3.23	4.08
Δ (k=0 $-$ k=10)		1.20	2.09	3.60	4.60

H3: prompt-level association. Within the attacked $k = 3$ condition the 25 traced prompts split into 8 refusers and 17 compliers. The prompt-mean refusal-direction projection is computed per prompt, then compared between groups; see Table 3 and Figure 3.

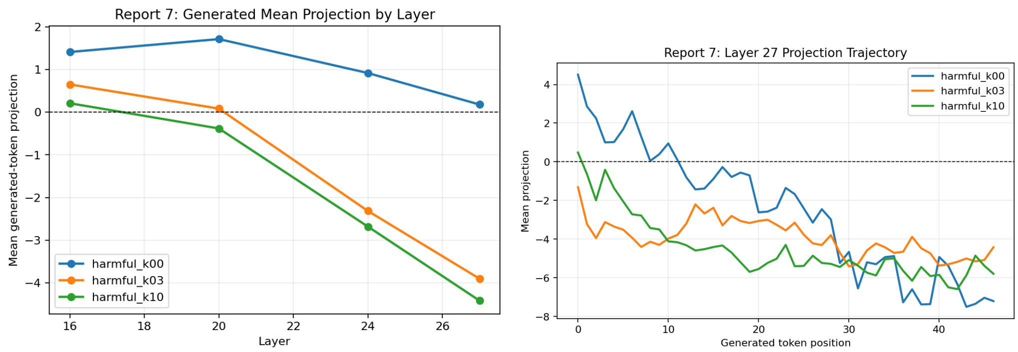

Figure 2: (left) Generated-token mean projection by layer for each condition; (right) layer-27 mean projection vs. generated-token position under $k = 0$ and $k = 3$. **Conclusion to draw:** the late-layer negative shift is concentrated at layers 24 and 27 and is sustained across the early portion of the response. H2 holds.

Table 3: Prompt-level association within $k = 3$ (held-out direction). Refusers ($n = 8$) versus compliers ($n = 17$); source: `outputs/report7/summaries/prompt_projection_label_differences.csv`. **Conclusion to draw:** refusers have less-negative late-layer projection than compliers, and the gap grows monotonically with depth (+0.93, +1.39, +1.85 at layers 20, 24, 27). H3 is supported, with the same depth signature as H2.

Layer	$n_{\text{compliance}}$	n_{refusal}	Mean (compliance)	Mean (refusal)	Refuser − Complier
20	17	8	−0.12	+0.81	+0.93
24	17	8	−2.61	−1.21	+1.39
27	17	8	−4.29	−2.44	+1.85

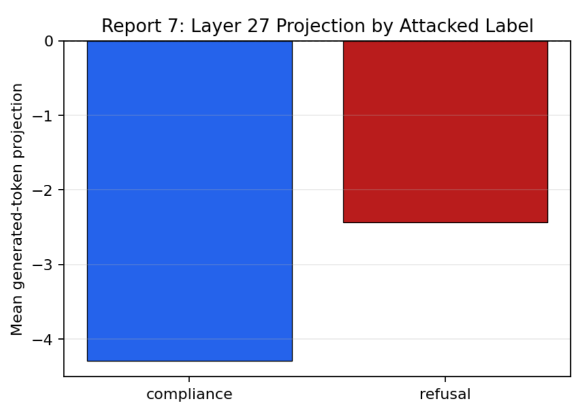

Figure 3: Mean layer-27 generated-token projection by attacked behavioral label. **Conclusion to draw:** prompts that still refuse under attack have less-negative late-layer projection than prompts that comply (refuser−complier gap +1.85 at layer 27). This is the prompt-level shadow of H2 and supports H3.

5 Cross-Condition Patching Results (H4)

The patching block uses a clean $k = 0$ source state from the same prompt and replaces the refusal-direction component at one generated-token position in the attacked $k = 3$ forward pass. Grid: layers $\{16, 24, 27\} \times$ target positions $\{0, 1, 3, 5\} \times 25$ prompts $= 300$ records. The Report 6 patching design used $k = 3$ same-condition sources whose layer-27 source projection was already -0.458 (attacked-into-attacked); Report 7 instead injects an unattacked source whose layer-27 projection is $+12.98$. The intervention should therefore have substantially more headroom to restore refusal.

Table 4: Cross-condition patching results ($k = 0$ source $\rightarrow k = 3$ target). Each cell is 25 prompts; bootstrap 95% confidence intervals are computed by resampling per-prompt outcomes ($B = 2000$). Source: `outputs/report7/summaries/cross_condition_patching_stats.csv`. **Conclusion to draw:** no cell restores refusal at a rate above 0.08 and every CI overlaps zero. Two cells *lose* refusal at a higher rate than they restore it (layer 24 / tt=1: 4 lost vs. 0 restored; layer 27 / tt=1: 2 lost vs. 1 restored). McNemar's exact test cannot fire in any cell because the discordant counts are < 5. H4 is not supported under this intervention setting.

Layer	tt	Restored	Lost	Restoration rate	95% CI
16	0	1	0	0.04	$[0.00, 0.12]$
16	1	0	2	0.00	$[0.00, 0.00]$
16	3	1	0	0.04	$[0.00, 0.12]$
16	5	0	0	0.00	$[0.00, 0.00]$
24	0	2	2	0.08	$[0.00, 0.20]$
24	1	0	4	0.00	$[0.00, 0.00]$
24	3	0	1	0.00	$[0.00, 0.00]$
24	5	1	0	0.04	$[0.00, 0.12]$
27	0	0	0	0.00	$[0.00, 0.00]$
27	1	1	2	0.04	$[0.00, 0.12]$
27	3	1	0	0.04	$[0.00, 0.12]$
27	5	1	0	0.04	$[0.00, 0.12]$

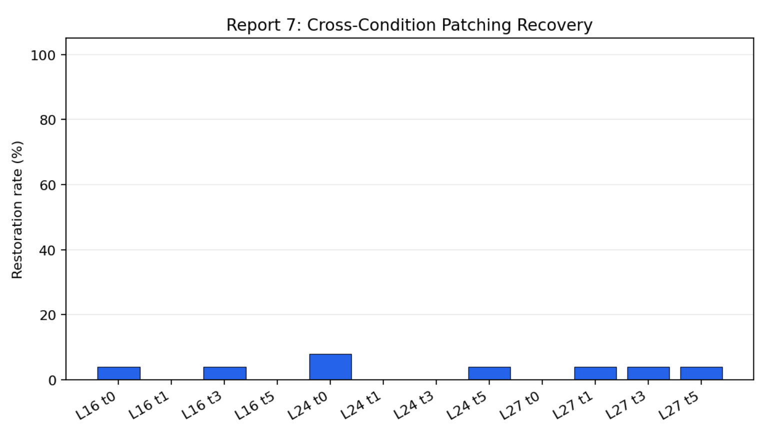

Figure 4: Cross-condition patching restoration rate by (layer, target position). **Conclusion to draw:** no cell exceeds 0.08; every 95% CI includes zero. H4 is not supported in this intervention setting, even with a clean unattacked source.

The Report 6 hypothesis that the same-condition $k = 3$ source was the limiting factor in H4 was a reasonable read of the layer-27 source projection (-0.458). However, replacing it with a clean source whose layer-27 projection is $+12.98$ does not change the picture: restoration remains within sampling noise of zero. The Report 6 patching null was *not* primarily a source-quality artifact.

6 Additive Direction Intervention (H5, H6)

Cross-condition patching transfers an entire source state; the additive intervention is tighter — it adds $\alpha \cdot \hat{d}_\ell$ to the attacked residual at one generated-token position, with $\hat{d}_\ell$ the unit-norm refusal direction at layer ℓ. The grid is layers $\{16, 24, 27\}$ × target positions $\{0, 1, 3\}$ × $\alpha \in \{1, 2\}$ × 25 prompts × three direction types (refusal; random unit vector, 5 seeds; orthogonal-to-refusal unit vector, 5 seeds). Total: 4,950 attacked-condition generations. The random and orthogonal direction types define the noise floor for a single-position residual edit at this site.

Table 5: Mean restoration rate (averaged across seeds for random and orthogonal) by direction type at the late layers (24, 27), pooled across target positions and α values. Source: `outputs/report7/interventions/additive_direction_seed_aggregated.csv`. **Conclusion to draw:** the refusal direction restores refusal in *zero* of 12 late-layer cells (300 of 300 prompts unchanged); random and orthogonal controls average ≈0.5% across cells. The refusal direction is at or below the noise floor: H5 is strongly null at L24/L27, and H6 is consistent with that null (refusal is no more effective than a random unit vector).

Direction type	Mean restoration rate	Max	n cells
Refusal	0.000	0.00	12
Random	0.005	0.02	12
Orthogonal	0.004	0.01	12

Table 6: Per-cell additive intervention restoration rate at the late layers (refusal direction only). All 12 cells return zero. Per-seed and combined CIs are in `outputs/report7/summaries/additive_direction_stats.csv`; McNemar cannot fire in any cell (discordant = 0). **Conclusion to draw:** even with the strongest available causal lever (the empirically extracted refusal direction, added at the cleanest restoration target position 0 with $\alpha = 2$), no cell flips behavior at the late layers.

Layer	tt	$\alpha = 1.0$	$\alpha = 2.0$
24	0	0.00	0.00
24	1	0.00	0.00
24	3	0.00	0.00
27	0	0.00	0.00
27	1	0.00	0.00
27	3	0.00	0.00

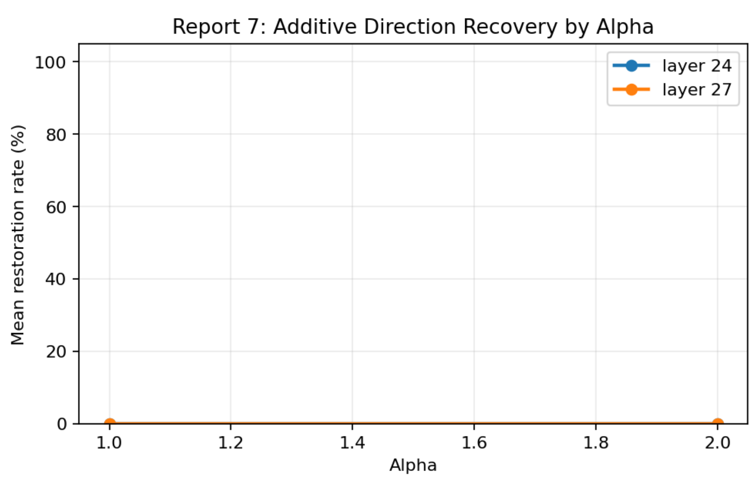

Figure 5: Additive intervention restoration rate by α, broken out by direction type. **Conclusion to draw:** no α moves the refusal direction above the random/orthogonal noise floor at the late layers. The intervention does not appear to depend on α in the 1.0–2.0 range under this design.

H6 positive control on benign prompts. The same additive grid is run on 25 benign Alpaca prompts at $k = 0$. The *intended* reading of restoration here is *false-refusal induction*: if the refusal direction is causally active in a non-attacking context, adding $\alpha \cdot \hat{d}_\ell$ to a benign residual should make the model refuse a benign request. We observe *zero* false-refusals across all 54 benign cells (3 layers $\times$ 3 target positions $\times$ 2 α $\times$ 3 direction types). The intervention has no measurable behavioral effect on benign prompts. Combined with the harmful null in Table 5, this indicates the additive intervention at this site (one generated-token position) is essentially causally inert in either direction.

7 Cross-Report Comparison

Table 7: Cross-report comparison of intervention restoration rates, pooled across cells at each layer. Source: `outputs/report7/summaries/report6_vs_report7_intervention_comparison.csv`. **Conclusion to draw:** fixing the source quality (R6 same-condition attacked source $\rightarrow$ R7 cross-condition clean source) and broadening to a stronger causal lever (additive refusal-direction injection at exactly the right site) does not change the qualitative conclusion: the late-layer refusal-direction signal at a single generated-token position cannot be intervened on to restore refusal under attack.

Method	Layer	Cells (sum of restored / sum of n)	Restoration rate
R6 same-condition attacked-source patching	16	0/30	0.00
R6 same-condition attacked-source patching	24	0/30	0.00
R6 same-condition attacked-source patching	27	0/30	0.00
R7 cross-condition baseline-source patching	16	2/100	0.02
R7 cross-condition baseline-source patching	24	3/100	0.03
R7 cross-condition baseline-source patching	27	3/100	0.03
R7 additive refusal $\alpha = 1.0$	24	0/75	0.00
R7 additive refusal $\alpha = 1.0$	27	0/75	0.00
R7 additive refusal $\alpha = 2.0$	24	0/75	0.00
R7 additive refusal $\alpha = 2.0$	27	0/75	0.00

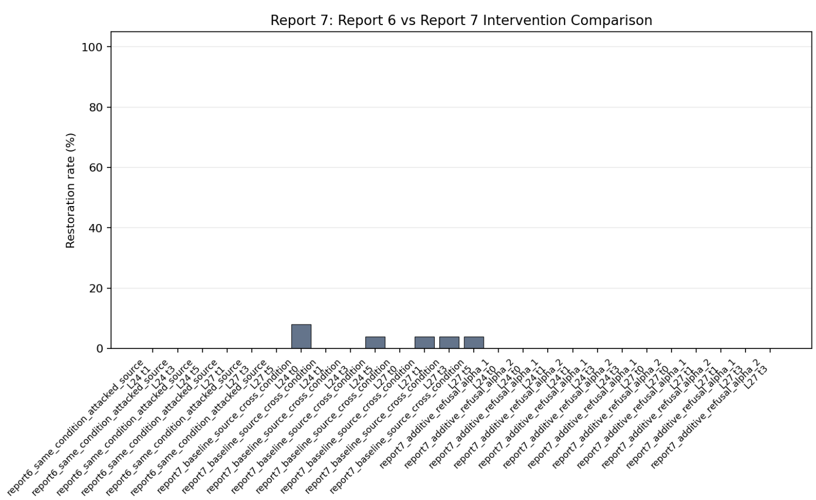

Figure 6: Restoration rate across intervention methods. **Conclusion to draw:** restoration stays near zero across three increasingly clean intervention designs (R6 attacked-source $\rightarrow$ R7 clean-source $\rightarrow$ R7 direct additive injection). The null is robust to source-quality, direction-magnitude, and direction-specificity controls.

8 Conclusions and Threats to Validity

Status of each hypothesis (Report 7 evidence only). H1, H2, H3 are supported. H4, H5, H6 are not supported in the direction they were originally written.

Table 8: Hypothesis status after Report 7.

H	Status	Comment
H1	Supported	0.92 $\rightarrow$ 0.32 at $k = 3$; 0.36 at $k = 10$.
H2	Supported (held-out)	Monotone deltas 0.76, 1.63, 3.23, 4.08 at L16/20/24/27.
H3	Supported	Refuser $-$ complier gap $+0.93$, $+1.39$, $+1.85$ at L20/24/27.
H4	Not supported	All cells ≤ 0.08 restoration, CIs overlap 0.
H5	Not supported (strongly)	Refusal direction restores 0 of 300 prompts at L24/L27.
H6	Mixed (see comment)	Refusal vs. controls indistinguishable at zero; benign positive control: 0 false-refusals.

Information loss vs. active suppression. The Report 5 design treated information loss and active suppression as two competing mechanistic accounts of the attack. The discriminator was: if a clean intervention restores refusal, the information-loss account is supported; if it does not, then the late-layer signal is more likely to be a downstream readout or to be overridden by other computations. Report 7 ran two cleaner interventions with controls. Both fail at the late layers, even though the observational signal at those same layers is unambiguous: the refusal direction is robust under a held-out extraction set (H2), tracks prompt-level behavior under attack (H3), and additive injection at early generated positions does nothing distinguishable from random or orthogonal controls (H5/H6).

The safest interpretation is therefore not simple information loss. The late-layer refusal-direction projection appears to be a real and predictive readout of the model's attacked state, but under the intervention granularity tested here it is not a direct behavioral knob. One plausible explanation is active suppression or "locking-in": by the time generation begins, the prefill has already shaped the model state enough that a single-position residual edit in layers 24/27 is too local and too late to recover refusal. This does not prove the full suppression mechanism, because multi-position edits, earlier residual interventions, or attention-routing interventions were not tested. It does, however, rule out the simpler account that restoring or adding the late-layer refusal-direction component at one generated-token position is sufficient to recover refusal. The same null also holds in the benign positive control: adding the refusal direction to a benign residual does not induce false-refusal, so the direction is not a free causal knob on behavior regardless of context.

Threats to validity.

- *Classifier validity.* A stratified spot-check of 32 outputs was independently re-labelled with a secondary classifier (Claude Opus 4.7); auto-vs-secondary agreement is 87.5% (28/32) overall and 100% on benign $k = 0$ and harmful $k = 0$. The three flips are all in attacked conditions: two cases where the model pivots to a safer related topic ("soft refusal via redirect") and one case where the disclaimer contained the word "illegal" and produced a false-positive refusal label. This means the Report 7 refusal-rate estimates at $k = 0$ are very reliable, while the attacked refusal counts may be off by $\pm$1–2 prompts on each side. Spot-check source data: `outputs/report7/classifier_spot_check/`.
- *Intervention scope.* The additive intervention is one-time at a single generated-token position. Multi-position or attention-routed interventions are not tested and may produce different results.
- *Sample size.* $n = 25$ prompts per cell; bootstrap CIs are wide and McNemar cannot fire in any patching cell. A larger sample could detect small effects below this resolution.
- *Single model, single attack.* Only Llama-3.1-8B-Instruct under the prefilling family is tested. The active-suppression reading does not necessarily transfer to suffix attacks or to other aligned models.
- *Layer-specific magnitudes are not comparable across layers.* The mean projection at layer 27 is on a different scale than at layer 16, so cross-layer numeric comparisons (e.g., the delta gradient) describe the relative direction of the effect, not its raw magnitude.

What this means for the project. The most informative outcome was that two cleaner experiments failed in the same direction as Report 6's single, methodologically weaker one. The observational evidence (H1–H3) is strong, the causal evidence at the single-position generation-side intervention site is reliably null, and the controls (random/orthogonal/benign) confirm the null is not a direction-quality artifact. The refusal direction is a real and prompt-level-predictive readout, but at the intervention granularity tested it is not the causal lever for refusal behavior under prefilling. The remaining mechanistic question — where exactly the locking-in happens, and whether attention-routing or earlier residual edits can overturn it — is a natural next step beyond the scope of this report.

References

[1] Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. *arXiv preprint arXiv:2406.11717*, 2024.

[2] Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

[3] Xiangyu Qi, Ashwinee Panda, Kaifeng Lyu, Xiao Ma, Subhrajit Roy, Ahmad Beirami, Prateek Mittal, and Peter Henderson. Safety alignment should be made more than just a few tokens deep. In *International Conference on Learning Representations (ICLR)*, 2025.

[4] Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. Stanford Alpaca: An instruction-following LLaMA model. GitHub repository, `https://github.com/tatsu-lab/stanford_alpaca`, 2023.

[5] Andy Zou, Zifan Wang, J. Zico Kolter, and Matt Fredrikson. Universal and transferable adversarial attacks on aligned language models. *arXiv preprint arXiv:2307.15043*, 2023.